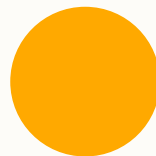




Intro to Javascript

Reese Hatfield



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Javascript

- All of our websites have been static
- They don't actually do anything
- Except render pure text
- How can we make our pages dynamic?





Javascript

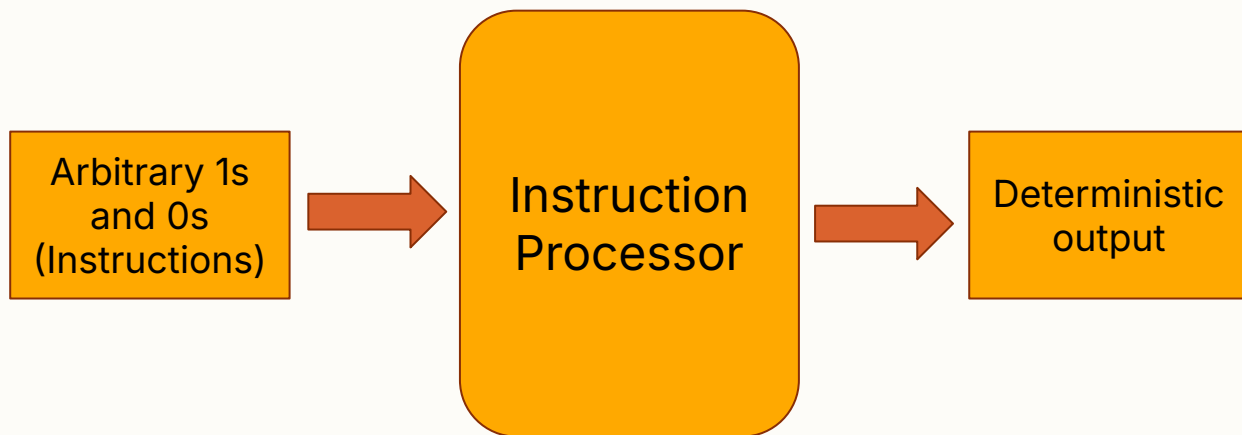
- Javascript
- Programming Language
- Different from everything we have used so far
- Let's think back a bit more

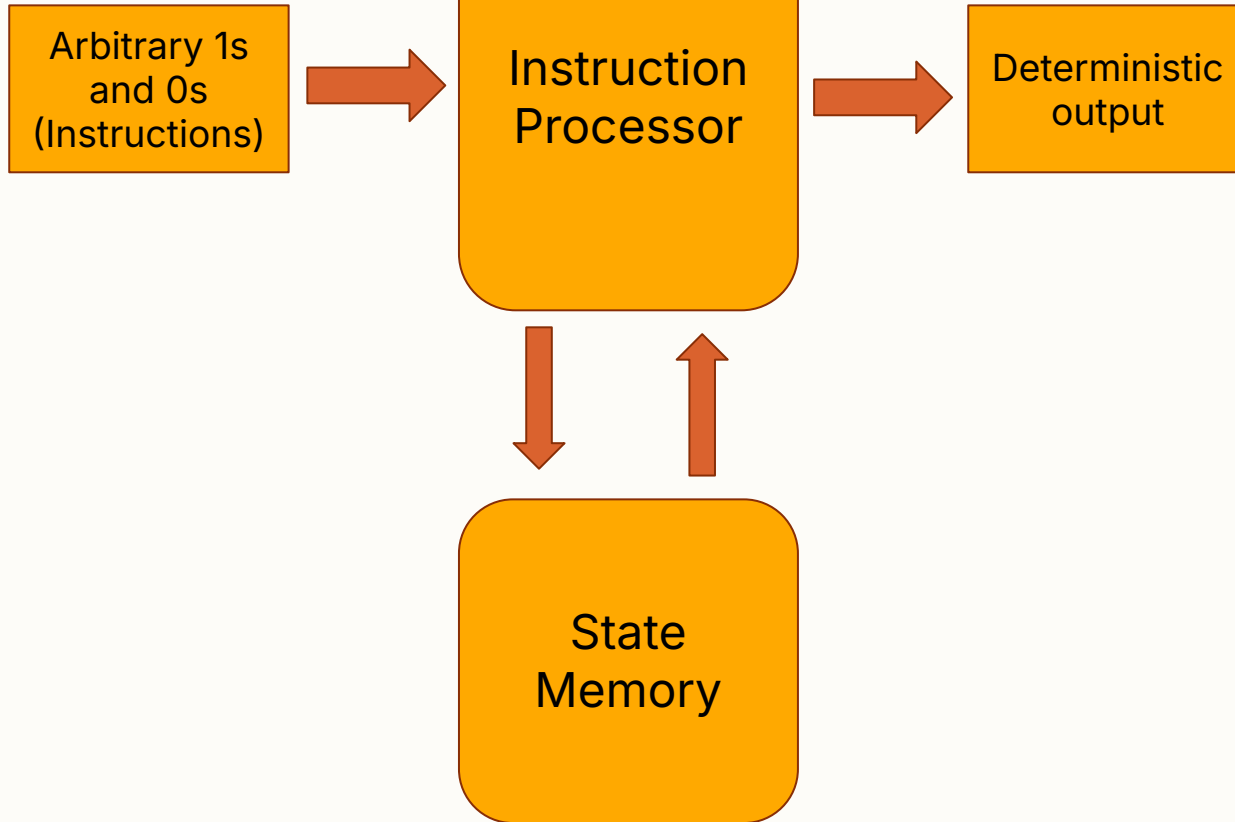




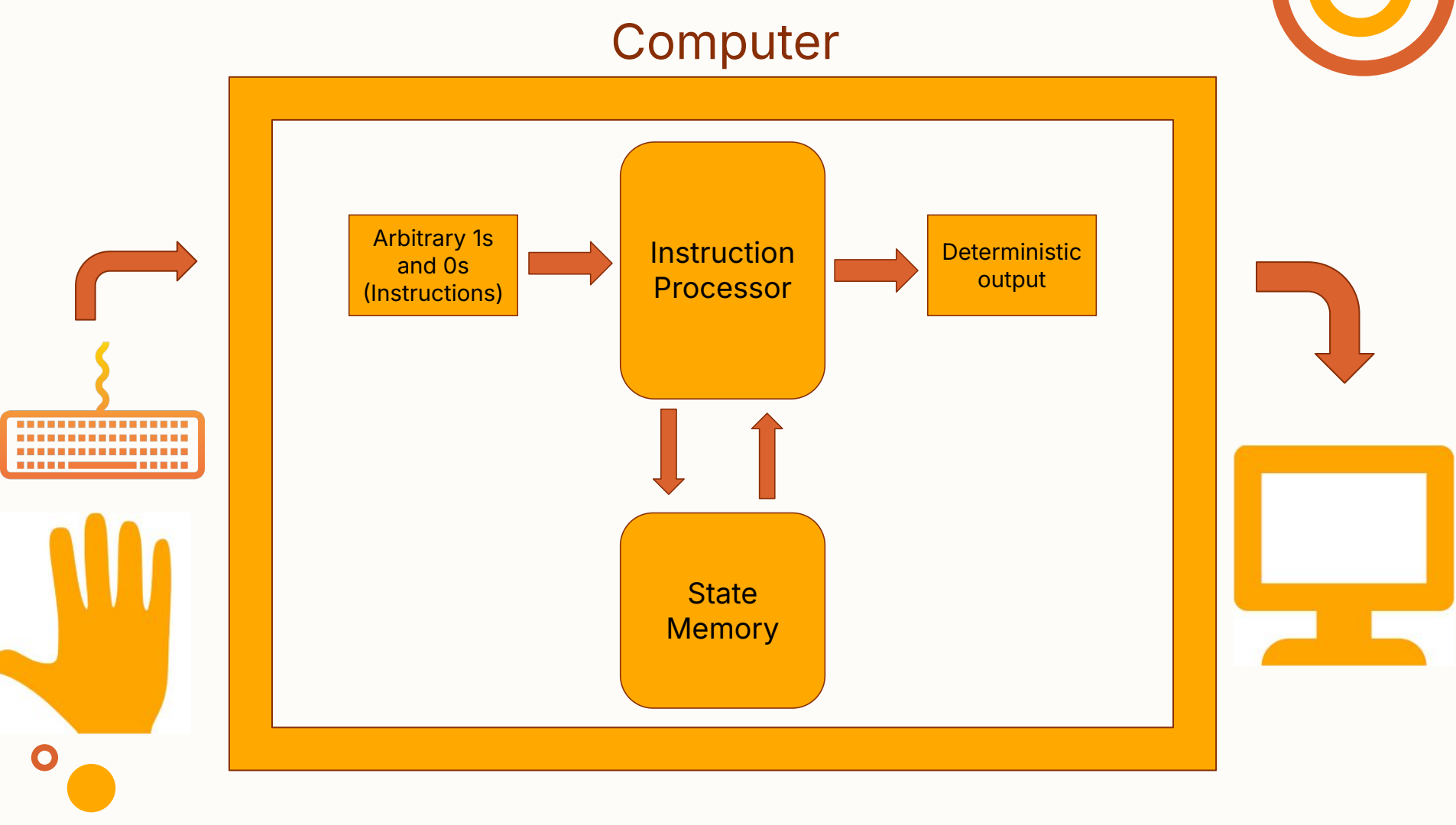
Abstraction

- Built hardware
- That take arbitrary instructions





Computer





Javascript

- HTML just renders text
 - That pretty much all it can do
- Javascript is an entire programming language
- You can do anything
 - Anything by definition





Javascript

- Programming languages let you write arbitrary code
- You could write the entire language definition in it
- Literally and formally anything
- This is an idea older than computers



Javascript

- Alan Turing
- Formulated the idea of "Turing Completeness"
- Anything computable can be simulated via a Turing Machine



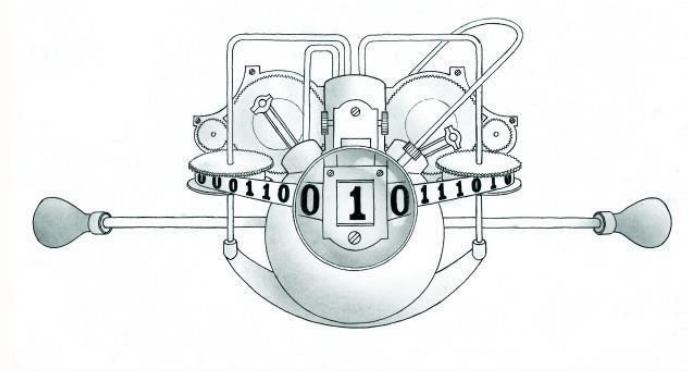
Javascript

- Turing machine is a theoretical device
- Has a tape of "cells"
 - Cell has a 1/0
- Has a reader
- Reader follows instructions



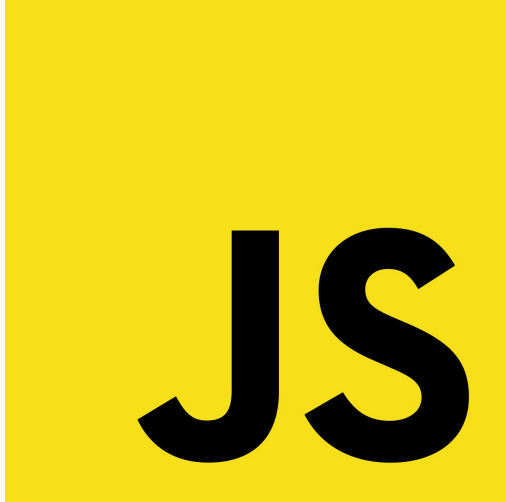
Javascript

- What cool about this?
- Published in **1936**
- Well before the modern concept of a computer existed
- A turing machine is basically a programming language



Javascript

- Javascript (like most all programming languages) is **turing complete**
- So it can be simulated with a turing machine
- Therefore, it can compute anything

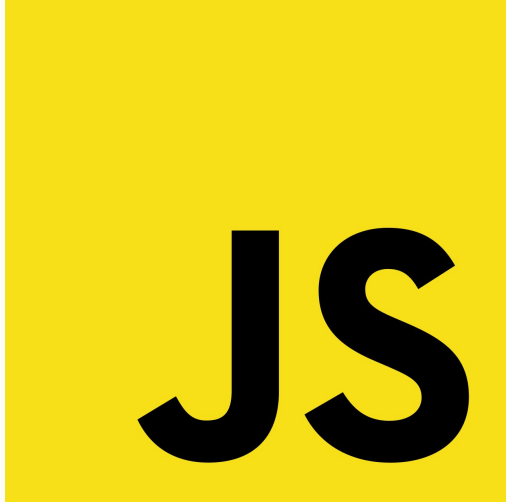
A yellow square containing the letters 'JS' in a bold, black, sans-serif font, representing the JavaScript logo.

JS



Javascript

- Javascript is built into the browser
- "Javascript Runtime"
- Meaning:
 - Has a lot of built ins
 - Ways of manipulating web pages
 - Dynamically

A yellow square containing the letters 'JS' in a bold, black, sans-serif font.

JS





Javascript

- Similar to HTML, JS is very particular about your "syntax"
- Has to be typed according to an exact style
- We can add this syntax with a
- **<script> ... </script>** tag





Javascript

- **alert()**
- Alert is the first way we will talk about to make our website more interactive
- Alert is a function (more of this later)
- We can "call it" with any data we want
alert("Hello"); alert("Reese");





Javascript

- Everything in our script tag will run top to bottom
- Right now we just have a call to an alert
- We can pass many things into alert
- These things have different types





Javascript

- **alert("Hello world");**
 - Used quotes
 - This is a String
 - We talked about these a long time ago
 - "String of characters"
- **alert("CS-1150");**





Javascript

- **alert(45);**
 - Numbers
 - Does not use a quote
 - Can have decimals
 - **alert(3.14159);**
 - Floating Point numbers!
 - Same IEEE defined encoding as before
- 



Javascript

- **alert(true);**
 - **alert(false);**
 - True or false
 - "Boolean" values
 - Not super useful yet
 - However, booleans exist as their own type we can use to control our program
- 



Javascript

- Javascript has more types than this
 - Can even define your own
- But those three are all we will need for this class
- number → Numerical data
- string → textual data
- boolean → true/false values





Javascript

- Types are an important concept in programming languages
- JS kinda fudges the rules a bit
- Plays more fast and loose than most languages
- Let's think about the different types of things we can alert() with





Javascript

- Alert really “takes over” the page
- If we want to write HTML directly, we can use the DOM
 - Document Object Model
- **document.write(“hello world”);**
- We can write arbitrary html here
- Let’s play around with that a bit





Javascript

- How can we make our program do a little more?
 - We need to cover a few basic programming concepts first
 - We already how we can tell our user something with **alert()** or **write()**
 - Few more concepts
- 



Javascript

- Variables
- Programming concept of a holder
- All variables:
 - Have a name
 - Contain some data*
 - Have a type (e.g. number, string, boolean, etc)
 - Have a declaration and an initialization





Javascript

- Anatomy of a variable
- **var** keyword in Javascript*

var favNumber; ← Declaration

...

favNumber = 12; ← Assignment

alert(favNumber); ← Com. w/ user





Javascript

- Declare with the **var** keyword
- Assign value with the = sign
 - Assignment operator
- There are other ways of declaring variables
 - Not important for us





Javascript

- Reassign Variables
- Code runs top to bottom

var favNumber; ← Declaration

favNumber = 12; ← Assignment

favNumber = 15; ← Re-assignment

alert(favNumber); ← Com. w/ user





Javascript

- String Concatenation
 - Combine values (strings) together
 - Use the + operator

var favWord; ← Declaration

...

favWord = "pizza"; ← Assignment

alert("Your # was: " + favWord);





Javascript

- Variable declaration and assignment
- Often done in 1 line of code

var favNumber = 15; ← Declare+Assign

alert(favNumber); ← Com. w/ user





Javascript

- We can do math if our types are numbers
- Supports common operations
 - +
 - -
 - *
 - /
- **Let's see that**





Javascript

- Add pi and age together
- Put the result into **var output**

```
var pi = 3.14;  
var age = 43;
```

```
var output = pi + age;  
alert(output);
```





Javascript

- We would like a way to gather input from the user
- To really make our website interactive
- Javascript also ships a built-in function for this
- **prompt()**





Javascript

prompt("message")

- Displays a prompt dialog
- Gives us back whatever the user types in
- Assign it to a variable

var answer = prompt("How are you?")





Javascript

- Using this variable
- We can now do things dynamically in our code

```
alert("You replied with " + answer)  
document.write("You said "+ answer)
```

- 
- Few catches



Javascript

- We need to consider our data types
- **prompt()** always replies with a string
- Meaning the + would do concatenation
 - Let's see that
- We might want our **string** as a **number**





Javascript

- **parseInt(x)**
- Parses a string into number

```
var x = prompt("Fav number?");
```

```
x = parseInt(x);  
var newNumber = x + 5;  
alert(newNumber);
```





Javascript

- Let's practice with this a bit
- Now that we can:
 - Print anything
 - Read in any input
- Called this input and output
 - Basic I/O





Javascript

- Let's make a BMI calculator
 - Classic example
- Let's first think about what all we would need
 - Height
 - Weight
 - Do some math?
 - Give some output



Javascript

- Formula
- Things to consider:
 - What units are we using
 - Can we convert?
 - Start simple
 - Break the problem down

$$\text{BMI} = \frac{\text{Weight in kilogram}}{(\text{Height in meter})^2}$$





Javascript

- Variables
- Ask: What things can change?
- Height
- Weight
- BMI
- Height Square?

$$\text{BMI} = \frac{\text{Weight in kilogram}}{(\text{Height in meter})^2}$$



Javascript

- Let's implement that
- If time permits
 - Let's convert units
- Important:
 - Communicate to the user what the input should be

$$\text{BMI} = \frac{\text{Weight in kilogram}}{(\text{Height in meter})^2}$$



Javascript

- More Practice
- Heights:
- The avg. Giraffe is 192 in. tall
- User enters:
 - Their height in feet, then inches
 - e.g 5, 10
- Determine how much taller the average giraffe is than the user

